FIITJEE
ALL INDIA TEST SERIES
JEE (Advanced)-2024
PART TEST – II
PAPER –1
TEST DATE: 10-12-2023

Time Allotted: 3 Hours **Maximum Marks: 180**

General Instructions:

- The test consists of total 51 questions.
- Each subject (PCM) has 17 questions.
- This question paper contains **Three Parts.**
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each **Part** is further divided into **Two Sections**: **Section-A & Section-B**.

Section – A *(01 –03, 18 – 20, 35 – 37):* This section contains **NINE (9)** questions. Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).

Section – A *(04 – 07, 21 – 24, 38 – 41):* This section contains **TWELVE (12)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.

Section – A *(08 – 11, 25 – 28, 42 – 45):* This section contains **TWELVE (12)** Matching List Type Questions. Each question has **FOUR** statements in **List-I** entries (P), (Q), (R) and (S) and **FIVE** statements in **List-II** entries (1), (2), (3), (4) and (5). The codes for lists have choices (A), (B), (C), (D) out of which, **ONLY ONE** of these four options is correct answer.

Section – B *(12 – 17, 29 – 34, 46 – 51):* This section contains **EIGHTEEN (18)** numerical based questions. The answer to each question is a **NON-NEGATIVE INTEGER VALUE**.

MARKING SCHEME

Section – A (One or More than One Correct): Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+4	If only (all) the correct option(s) is (are) chosen;
Partial Marks	:	+3	If all the four options are correct but ONLY three options are chosen;
Partial marks	:	+2	If three or more options are correct but ONLY two options are chosen and both of which are correct;
Partial Marks	:	+1	If two or more options are correct but ONLY one option is chosen and it is a correct option;
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered);
Negative Marks	:	–2	In all other cases.

Section – A (Single Correct): Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+3	If ONLY the correct option is chosen.
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered);
Negative Marks	:	–1	In all other cases.

Section – B: Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+4	If ONLY the correct numerical value is entered at the designated place;
Zero Marks	:	0	In all other cases.

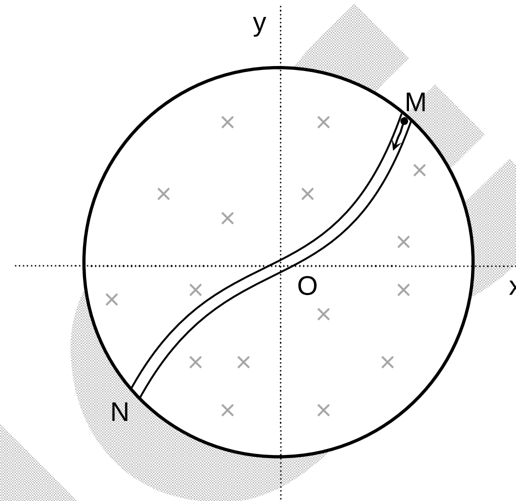
Physics

PART – I

SECTION – A
(One or More than one correct type)

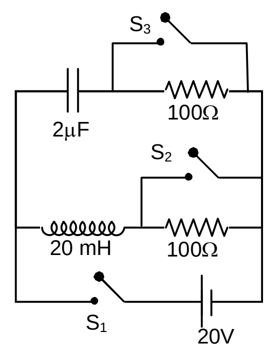
*This section contains **THREE (03)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).*

1. Magnetic field is increasing in a circular region of radius 2 m at a rate 30 Tesla/sec. O is the centre of circular region which is taken as origin. A smooth tunnel is formed in this plane which is represented as $y = \pm\left(\sqrt{3}\right)x^2$. A charge particle q enters into the tunnel at M with a kinetic energy E_1 and leaves the tunnel at N with kinetic energy E_2. Mark the correct statement.



(A) From M to O kinetic energy will decrease
(B) From O to N kinetic energy will increase
(C) $E_1 - E_2$ will be zero
(D) $E_1 - E_O$ will be $5\sqrt{3}$ J
(E_O is kinetic energy of charge particle at origin)

2. In the given circuit switch S_2 and S_3 are open and S_1 is closed for long time such that a constant current is flowing through the battery. Now at t = 0, S_2 and S_3 are closed and simultaneously S_1 is opened. Mark the correct statement(s).



(A) Maximum energy stored in electric and magnetic field will be 800μJ
(B) Charge in capacitor as a function of time is $Q = 40\sin\left(\frac{10^4 t}{2} + \frac{3\pi}{4}\right)\mu C$
(C) Maximum current in inductor at any time t > 0 will be $\frac{\sqrt{2}}{5}$ A
(D) Time when energy stored in magnetic field = energy stored in electric field first time will be $\frac{3\pi}{4} \times 10^{-4}$ sec

3. A body is radiating energy. E is the emissive power of a body and λ_m is the wave length corresponding to which spectral emissive power is maximum and T is temperature in Kelvin scale. A graph $\log_e T$ versus $\log_e \lambda_m$ and $\log_e E$ versus $\log_e T$ are shown in the figures. Mark the correct statement(s).

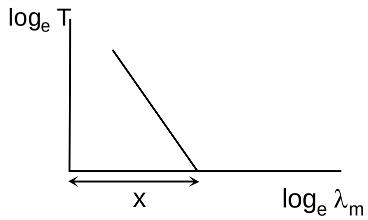


Figure A

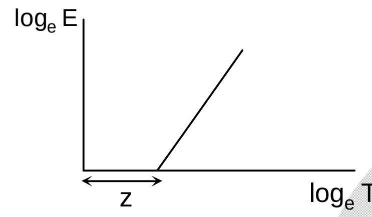


Figure B

(A) Slop of line in figure A, is –1
(B) Slop of line in figure B is 4
(C) Weins constant $\log_e x$
(D) the value of wein's constant is e^x

SECTION – A
(One Options Correct Type)

*This section contains **FOUR (04)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.*

4. Two moles of an ideal diatomic gas expands and follows a thermodynamics process $P = A - BV^2$, here A and B are positive constant. Mark the correct statement regarding the process.

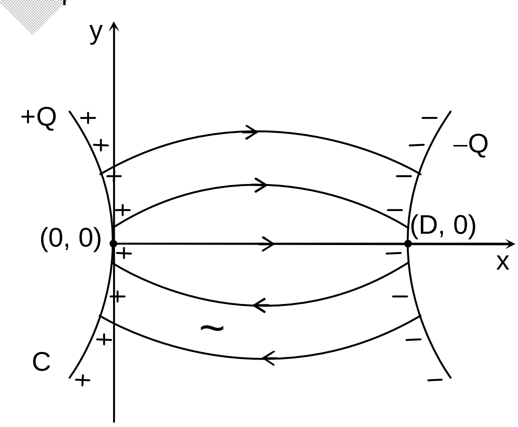
(A) Temperature is minimum at $V = \sqrt{\frac{A}{3B}}$

(B) Temperature is maximum at $V = \sqrt{\frac{A}{3B}}$

(C) Temperature first decreases then increase

(D) The rate of change of temperature increases with expansion.

5. A system of two conductors carry equal and opposite of charges Q and –Q respectively. The electric field thus created is non uniform as shown in the figure. The electric field with x-axis varies as $E_x = QA\varepsilon_0\left(1+\frac{x^2}{B}\right)$, where A and B are constant. Minimum separation between conducting surface is D on x-axis. Then,



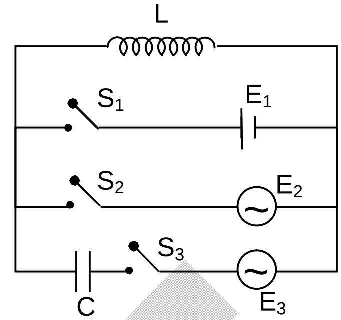
(A) Equivalent capacitance of system will be $\frac{3B}{A\varepsilon_0 D(3B - D^2)}$

(B) Surface charge density near origin is $\frac{QA\varepsilon^2}{2}$

(C) Equivalent capacitance of system will be $\frac{3B}{A\varepsilon_0 D(3B + D^2)}$

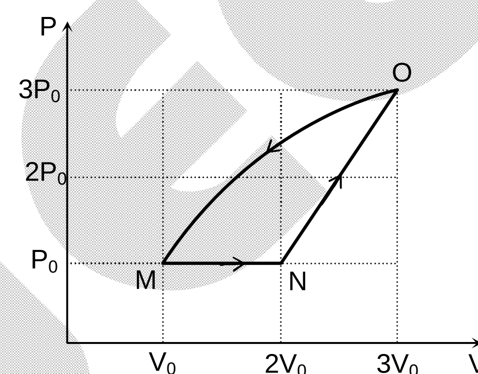
(D) Surface charge density near origin is $\frac{QA\varepsilon_0^2}{2}\left(1+\frac{D^2}{B}\right)$

6. In the given circuit
E_1 = 24 volt DC, E_2 = 24 volt AC 50 rad/s, E_3 = 12 volt AC 50 rad/s and C = $\frac{1}{50}$F .
With S_1 is closed and S_2, S_3 are open, a steady current 8 amp is observed. With S_1, S_3 open and S_2 closed current observed is 4.8 amp. Mark the correct answer.



(A) Inductance of coil is nearly is 0.8H
(B) Resistance of coil is 6Ω
(C) If S_3 closed with S_1 and S_2 open rms value of current in the circuit will be 2 A
(D) If S_3 is closed with S_1 and S_2 are open average power supplied by battery will be 24 watt

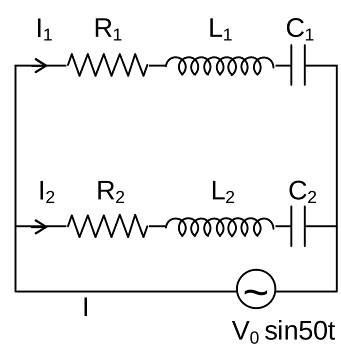
7. An ideal gas follows a cyclic process MNOM. In process MN heat absorbed by the gas is 200J where as in process OM internal energy of gas is decreased by 900J. The adiabatic constant for the gas will be
(A) 2/3
(B) 5/3
(C) 16/9
(D) 7/5



SECTION – A
(Matching List Type)

*This section contains **FOUR (04)** Matching List Type Questions. Each question has FOUR statements in **List-I** entries (P), (Q), (R) and (S) and FIVE statements in **List-II** entries (1), (2), (3), (4) and (5). The codes for lists have choices (A), (B), (C), (D) out of which **ONLY ONE** of these four options is correct answer.*

8. An alternating voltage of V = V_0sin 50t is applied acorss the above LCR circuit as shown in the figure. Now match the following list for various situations.



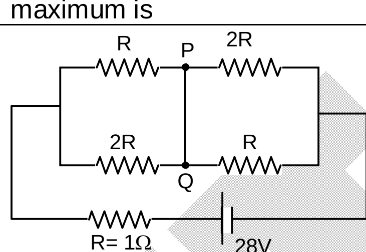
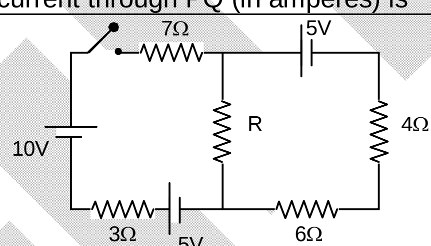
	List –I		List –II
(P)	If L_1 = 4H and C_1 = 200 μF, R_1 = 100 Ω and L_2 = 4H, C_2 = 100 μF , R_2 = 100 Ω	(1)	Phase difference between I_1 and I_2 will be π/2
(Q)	If L_1 = 2H and C_1 = 200 μF, R_1 = 100 Ω and L_2 = 4H, C_2 = 100 μF , R_2 = 100 Ω	(2)	Phase difference between I_1 and I_2 will be π/4
(R)	If L_1 = 8H and C_1 = 200μF, R_1 = 100$\sqrt{3}$ Ω and L_2 = 2H, C_2 = 50 μF , R_2 = 100$\sqrt{3}$ Ω	(3)	Phase difference between I_1 and I_2 will be 2π/3

(S)	If L_1 = 4H and C_1 = 200 μF, R_1 = 100 Ω and L_2 = 4H, C_2 = 50 μF , R_2 = 200 Ω	(4)	I_1 and I_2 will be in same pahse
		(5)	Phase difference between I_1 and I_2 will be $\pi/3$

The correct option is:

(A) (P) $\to$ (2) (Q) $\to$ (4) (R) $\to$ (3) (S) $\to$ (1)
(B) (P) $\to$ (2) (Q) $\to$ (4) (R) $\to$ (5) (S) $\to$ (1)
(C) (P) $\to$ (5) (Q) $\to$ (4) (R) $\to$ (3) (S) $\to$ (1)
(D) (P) $\to$ (2) (Q) $\to$ (4) (R) $\to$ (3) (S) $\to$ (5)

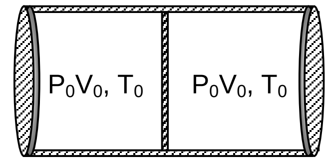
9. Match each entry in **List-I** to the correct entries in **List-II.**

	List –I		**List –II**
(P)	Magnitude of electric field (in V/m) at $\left(\frac{1}{\sqrt{6}}, \frac{1}{\sqrt{8}}, \frac{1}{\sqrt{10}}\right)$ if potential is given by $V = -\frac{1}{x} - \frac{1}{y} - \frac{1}{z}$ volt, is	(1)	4
(Q)	The charge flowing through a resistance 6Ω varies with t as $Q = 4t - 2t^2$. The total heat produced (in joules) across the resistance till the instantenous power is maximum is	(2)	10
(R)	 Wire PQ has negligible resistance the current through PQ (in amperes) is	(3)	5
(S)	 Value of R (in Ω) for which maximum power is generated across R when switch is closed will be	(4)	$10\sqrt{2}$
		(5)	32

The correct option is:

(A) (P) $\to$ (4) (Q) $\to$ (1) (R) $\to$ (5) (S) $\to$ (2)
(B) (P) $\to$ (4) (Q) $\to$ (3) (R) $\to$ (5) (S) $\to$ (1)
(C) (P) $\to$ (4) (Q) $\to$ (5) (R) $\to$ (3) (S) $\to$ (1)
(D) (P) $\to$ (4) (Q) $\to$ (5) (R) $\to$ (1) (S) $\to$ (3)

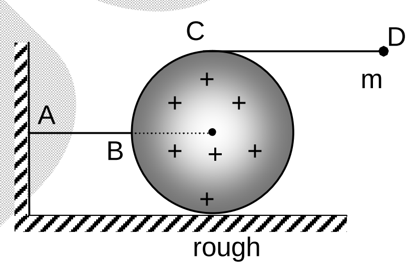
10. A thermally insulated cylinder is divided by a movable insulated piston in two equal chambers. Each chamber contains one mole of monoatomic ideal gas at P_0, V_0 and T_0. A coil is burnt in the left chamber due to which left chamber absorbs heat and expands pushing the piston towards right. In mechanical equilibrium pressure in right chamber is $32P_0$. Now match the following.



List –I		List –II	
(P)	The product of P and V in left chamber	(1)	$\frac{9}{2}P_0V_0$
(Q)	The work done by gas in right chamber	(2)	$60P_0V_0$
(R)	The change in internal energy of the gas in left chamber	(3)	$\frac{177}{2}P_0V_0$
(S)	The heat absorbed by the gas in left chamber	(4)	$93P_0V_0$
		(5)	$-\frac{9}{2}P_0V_0$

The correct option is:
(A) (P) → (2) (Q) → (1) (R) → (3) (S) → (4)
(B) (P) → (3) (Q) → (2) (R) → (5) (S) → (4)
(C) (P) → (2) (Q) → (1) (R) → (4) (S) → (5)
(D) (P) → (2) (Q) → (1) (R) → (3) (S) → (5)

11. A particle of mass m = 2 kg, charge q is connected to topmost point of a uniformly charged sphere of radius R = 30 cm, mass M with the help of a string CD of length $\frac{4R}{3}$ which is horizontal in equilibrium. One more string AB connects the sphere to a vertical support to keep the sphere stationary. ($g = 10\ m/s^2$)



List –I		List –II	
(P)	Electrostatic force on the charge particle	(1)	$\frac{100}{3}N$
(Q)	Tension in string AB	(2)	$\frac{80}{3}N$
(R)	Frictional force acting on sphere	(3)	$\frac{400}{3}N$
(S)	If string AB is cut the value of Ma_{cm}. If point of contact does not slides just after the cut and a_{cm} is instantaneous acceleration of centre of mass of sphere	(4)	$\frac{160}{3}N$
		(5)	$\frac{800}{21}N$

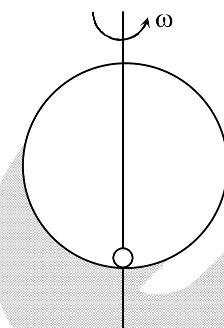
The correct option is:
(A) (P) → (3) (Q) → (2) (R) → (1) (S) → (4)
(B) (P) → (3) (Q) → (2) (R) → (5) (S) → (4)
(C) (P) → (1) (Q) → (4) (R) → (2) (S) → (5)
(D) (P) → (1) (Q) → (4) (R) → (2) (S) → (3)

SECTION – B

(Numerical Answer Type)

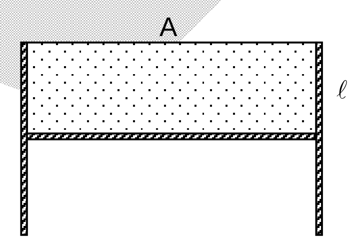
This section contains ***SIX (06)*** *Numerical based questions. The answer to each question is a* ***NON-NEGATIVE INTEGER VALUE****.*

12. Q = 2 mC charge is kept at the bottom of a smooth vertical hoop of radius 20 cm which is rotated with an angular velocity ω = 10 rad/s. The magnetic dipole moment of the hoop in amp-cm^2 (g = 10 m/s^2)

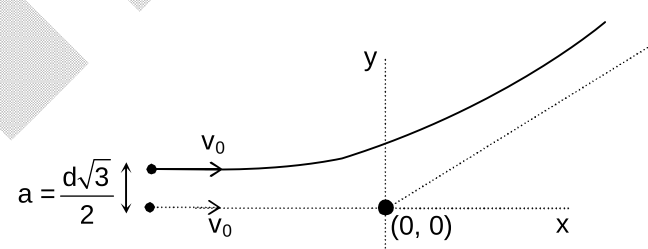


13. A DC ammeter and an AC thermal ammeter are connected in series in a circuit, when a DC current pass through the circuit, DC ammeter shows I_1 = 6A. When an AC flows through the circuit, the AC ammeter shows I_2 = 8A. What will the difference in final readings (in amp) of ammeters, if the DC and the AC flow simultaneously through the circuit?

14. A large conducting cylinderical container of area A contains a monoatomic gas inside a volume $A \times \ell$ such that the pressure of gas is same as atmospheric pressure (P_0). The bottom of this volume is a movable piston of 10N weight which is held fixed as shown in the figure. At t = 0 the piston is released such that it moves a distance x before coming to rest ($x << \ell$). The value of x is $\frac{k\ell}{P_0 A}$. Find the value of k.

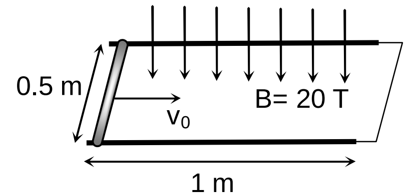


15. A charge particle Q is fixed at origin and another identical charge particle is given velocity $v_0\hat{i}$ along x-axis from large distance. The minimum separation between the two charges is observed d. If now the charge particle gives velocity $v_0\hat{i}$ along line $y = \frac{d\sqrt{3}}{2}$. The minimum separation between the charges is observed $\frac{kd}{4}$. Find the vlaue of k.



16. Two infinite long wires are having uniform linear charge density $+\lambda$ and $-\lambda$ respectively. They are parallel to z-axis and passes through x-axis at $x_1 = -a$ and $x_2 = +a$ respectively. The radius of an equipotential cylinder having potential $\frac{\lambda \ln(2)}{4\pi\varepsilon_0}$ is found $ka\sqrt{2}$. Find the value of k.

17. A horizontal rail road track of length 1 m is fixed on a horizontal surface. Resistance per unit length of the track is 5 Ω/m. Magnetic field B = 20 Tesla which is vertically downward. A conducting rod of mass 100 gm length 0.5 m and resistance 10 Ω is given velocity v_0 at one end of the rod. The other end of track is short circuited. Calculate the value of v_0 if rod just stop at other end. (given $\ell n2 = 0.7$)



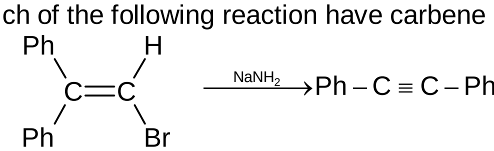
Chemistry

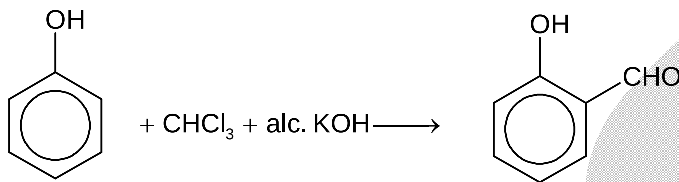
PART – II

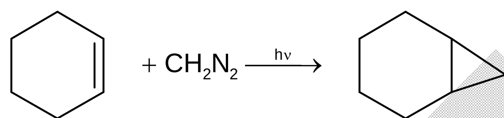
SECTION – A
(One or More than one correct type)

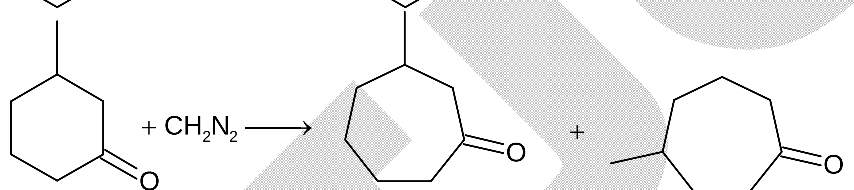
*This section contains **THREE (03)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).*

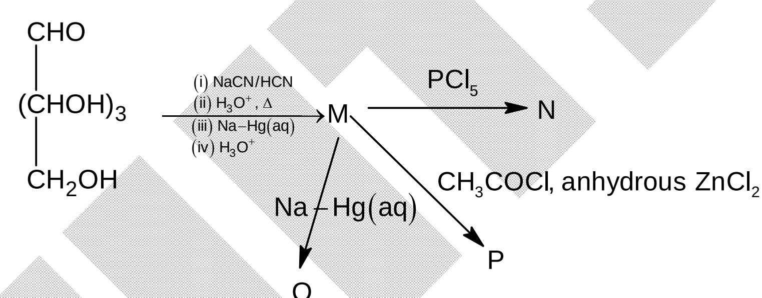
18. Which of the following reaction have carbene as intermediate?

(A) 

(B) 

(C) 

(D) 

19. 

Which of the following option(s) is/are correct?

(A) M is $CHO(CHOH)_4CH_2OH$

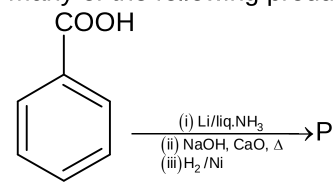
(B) N is a pentachloride

(C) P is a pentacetate

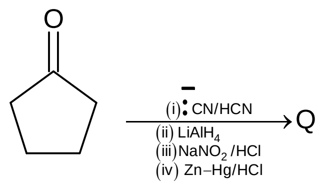
(D) Q is sorbitol $\left[CH_2OH(CHOH)_4CH_2OH\right]$

20. How many of the following product(s) is/are same

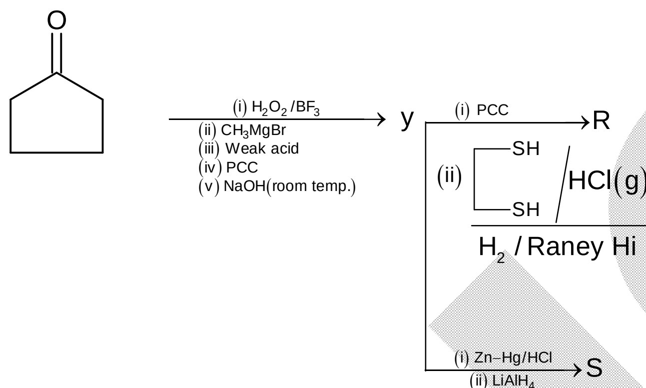
i)



ii)



iii)



(A) P
(B) Q
(C) R
(D) S

SECTION – A
(One Options Correct Type)

*This section contains **FOUR (04)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.*

21. In which of the following pairs, both species contain $\beta-D(+)$glucose as one of the component

(A) Starch, $\alpha-$Lactose
(B) Sucrose, Cellulose
(C) Starch, Maltose
(D) Cellulose, $\beta-$Lactose

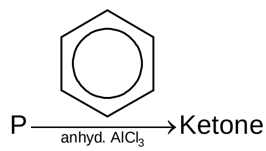
22. For which of the following Lassaigne's test will fail

Benzene, Hydrazine, $PhN_2^+Cl^-$, Urea, Phenyl chloride , Phenyl hydrazine, Hydrazoic acid

(A) Benzene, Urea, Phenyl chloride, Hydrazoic acid
(B) Hydrazine, Phenyl diazonium chloride, Phenyl hydrazine
(C) Benzene, Hydrazine, Phenyl diazonium chloride, Hydrazoic acid
(D) Hydrazine, Phenyl diazonium chloride, Urea, Phenyl chloride

23. $H_3C-\overset{\overset{\large O}{\|}}{C}-CH_3 \xrightarrow[\text{(ii) Tollen's reagent, (iii) conc. } H_2SO_4/\Delta\text{, (iv) } CH_2=C=O]{\text{(i) } SeO_2} P(\text{Product})$

Which of the following option is correct?

(i) 

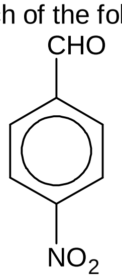
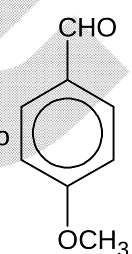
(ii) $P \xrightarrow[CH_3COONa]{PhCHO} \text{Cinnamic acid}$

(iii) $P \xrightarrow[\text{(ii) }\Delta]{\text{(i) } NH_3 \text{ (excess)}} Q \xrightarrow[\text{alc. KOH}]{Br_2} 1^o \text{ Amine}$

(iv) P is an acid

(A) Only (i) is correct
(B) Only (ii) is correct
(C) (i), (ii) and (iv) are correct
(D) (i), (ii) and (iii) are correct

24. Which of the following statement(s) are correct?

(i)  have greater rate of Cannizzaro reaction with respect to 

(ii) $Ph-\overset{\overset{\large O}{\|}}{C}-CH_2Cl$ gives faster S_N2 with $\ddot{I}^-$ w.r.t. CH_3Cl

(iii) Pinacol $H_3C-\underset{OH}{\overset{CH_3}{C}}-\underset{OH}{\overset{CH_3}{C}}-CH_3$ gives alkene after dehydration reaction with conc. H_2SO_4

(iv) Pinacol $H_3C-\underset{OH}{\overset{CH_3}{C}}-\underset{OH}{\overset{CH_3}{C}}-CH_3$ gives Pinacolone after rearrangement with conc. H_2SO_4.

(A) (i) & (ii)
(B) (i), (ii) & (iv)
(C) (ii) & (iv)
(D) (i), (ii) & (iii)

SECTION – A
(Matching List Type)

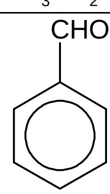
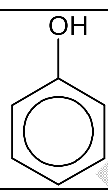
*This section contains **FOUR (04)** Matching List Type Questions. Each question has FOUR statements in **List-I** entries (P), (Q), (R) and (S) and FIVE statements in **List-II** entries (1), (2), (3), (4) and (5). The codes for lists have choices (A), (B), (C), (D) out of which **ONLY ONE** of these four options is correct answer.*

25. Match the following:

List – I		List – II	
(P)	Polyurethane	(1)	Natural polymer
(Q)	Glyptal	(2)	Biodegradable polymer
(R)	Gun cotton (Nitro cellulose)	(3)	Ester linkage
(S)	Glycine + Amino caproic acid $\xrightarrow{\text{condensation}}$ Polymer	(4)	Thermosetting
		(5)	Semi synthetic polymer

(A) P → 2; Q → 3; R → 5; S → 4
(B) P → 4; Q → 3; R → 5; S → 2
(C) P → 5; Q → 4; R → 2; S → 2
(D) P → 1; Q → 2; R → 4; S → 5

26. Match the following:

List – I		List – II	
(P)	CH_3CH_2OH	(1)	Tollen's test
(Q)		(2)	Fehling's solution test
(R)		(3)	Iodoform test
(S)	CH_3COOH	(4)	Br_2 water test
		(5)	$NaHCO_3$ test

(A) P → 3; Q → 2; R → 5; S → 4
(B) P → 3; Q → 1; R → 4; S → 5
(C) P → 2; Q → 2; R → 3; S → 4
(D) P → 4; Q → 3; R → 4; S → 5

27. Match the following:

List – I		List – II	
(P)	$Ph-CHO \xrightarrow{KCN}$	(1)	Cross Cannizzaro
(Q)	$CH_3CHO \xrightarrow{KCN}$	(2)	Mixed Claisen condensation
(R)	$PhCHO + HCHO \xrightarrow{NaOH}$	(3)	Benzoin condensation
(S)	$CH_3CHO + HCHO \xrightarrow{NaOH}$	(4)	Cyanohydrin formation
		(5)	At first Aldol reaction

(A) P → 3; Q → 4; R → 1; S → 5
(B) P → 4; Q → 4; R → 2; S → 1
(C) P → 4; Q → 4; R → 3; S → 2
(D) P → 3; Q → 4; R → 2; S → 1

28. Match the following:

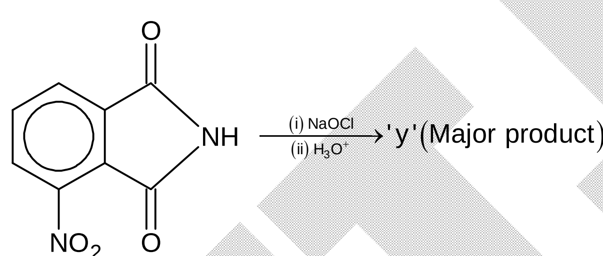
	List – I(Reaction)		List – II(Product)
(P)	$H_3C-\underset{\overset{\mid}{}}{\overset{OH}{\overset{\mid}{CH}}}-COOH \xrightarrow{\Delta}$ Product	(1)	Lactone
(Q)	$H_3C-\underset{\underset{OH}{\mid}}{CH}-CH_2COOH \xrightarrow{\Delta}$ Product	(2)	Lactide
(R)	$H_3C-\underset{\underset{OH}{\mid}}{CH}-CH_2-CH_2COOH \xrightarrow{\Delta}$ Product	(3)	Unsaturated acid
(S)	$H_3C-\underset{\underset{Cl}{\mid}}{CH}-CH_2-CH_2-COOH \xrightarrow{NaOH}$ Product	(4)	Optically active product
		(5)	Hydroxy acid

(A) P → 3; Q → 2; R → 5; S → 1
(B) P → 5; Q → 3; R → 4; S → 1
(C) P → 2; Q → 3; R → 1; S → 4
(D) P → 4; Q → 1; R → 2; S → 5

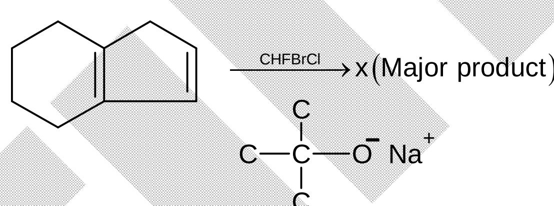
SECTION – B

(Numerical Answer Type)

This section contains ***SIX (06)*** *Numerical based questions. The answer to each question is a* ***NON-NEGATIVE INTEGER VALUE****.*

29. 

If 1 mole of reactant taken then what will be the amount (in gm) of product formed with 50% yield.

30. 

Molar mass of x is
[Use Atomic mass of C = 12, F = 19, Cl = 35, Br = 80]

31. $H_3C-CH_2-\underset{\underset{Br}{\mid}}{CH}-CH_3 \xrightarrow[\text{(ii) } CH_3Li]{\text{(i) }:PPh_3} A \xrightarrow{CH_3CHO} B$

1 mole

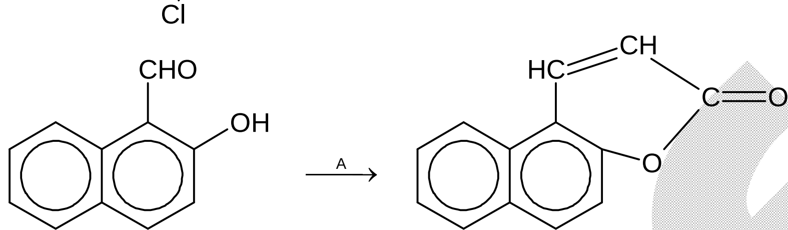
If yield of B is 75%. Then calculate the mass of B formed.

32. $Ph_3P + CCl_4 \longrightarrow A$

$$(C_2H_5)(H_3C)C{=}N{-}OH \xrightarrow{A} \underset{\text{(Major product)}}{B} + Ph_3P = O + CHCl_3$$

Number of 'H' atoms in product 'B' is x
Number of 'C' atoms in product 'B' is y
What is the value of x + y?

33. $Ph_3\overset{+}{P}\overset{-}{C}H_2 + (Cl)_2C{=}O \longrightarrow A$



Number of 'C' atoms in 'A' are x
Number of 'Cl' atoms in 'A' are y
Number of 'phenyl rings' in 'A' are z
Calculate x+ y + z = ?

34.

$$H-C\equiv C-OC_2H_5 \xrightarrow[0^{o}C]{C_2H_5MgBr} A \xrightarrow[0^{o}C]{Ph\overset{O}{\overset{\|}{C}}-CH_3} B \xrightarrow[BaSO_4]{H_2/Pd} \underset{(\text{Major})}{C} \xrightarrow[\text{reflux, 15 mins}]{0.1N\ HCl} Ph-\underset{CH_3}{\underset{|}{C}}{=}CH-CHO$$

Number of 'O' atoms in 'C' are x
Number of 'Carbon' atoms in 'C' are y
Number of 'Phenyl rings' in 'C' are z
What is the value of x + y + z

Mathematics

PART – III

SECTION – A

(One or More than one correct type)

*This section contains **THREE (03)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).*

35. Let $\left[\frac{\tan\alpha}{\sec\beta}+\frac{\sec\beta}{\tan\alpha}+\frac{\tan\beta}{\sec\alpha}+\frac{\sec\alpha}{\tan\beta}\right]^2 = p$ where α, β are distinct roots of the equation $24\sin^3 x - 26\sin^2 x + 9\sin x - 1 = 0$, then possible value(s) of p is/are

(A) $\left(\frac{35}{6}\right)^2\frac{25}{24}$
(B) $\left(\frac{63}{4}\right)^2\frac{1}{5}$
(C) $\left(\frac{143}{12}\right)^2\frac{49}{120}$
(D) $\left(\frac{63}{8}\right)^2\frac{83}{32}$

36. Let B, C, D represent foot of normals from point A to the curve P : $y^2 - 4x = 0$. Let S = {(B, C, D) : A lies on $x^2 + y^2 - 40x + 399 = 0$}. Let R = {(x, y) : (x, y) is centroid of ΔFGH; (F, G, H) $\in$ S}, then which of the following is/are TRUE?

(A) (11, 0) $\in$ R
(B) (12, 0) $\in$ R
(C) n(R $\cap$ S) = 0
(D) n(R $\cap$ S) = 2

37. Let S : $x^2 + y^2 = 4$ and L : x + y = 1. Let points where L intersects S be B and C. A is any variable point on the circle S. Locus of reflection of orthocentre of triangle ABC in BC be S′, then which of the following is/are TRUE?

(A) $\angle$BAC can be equal to $\tan^{-1}\left(\sqrt{7}\right)$
(B) $\angle$BAC can be equal to $\cot^{-1}\left(-\frac{1}{\sqrt{7}}\right)$
(C) Area bounded by S is more than 12 sq. units
(D) Number of points of intersection of S′ and L is 2

SECTION – A

(One Options Correct Type)

*This section contains **FOUR (04)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.*

38. Let H be a hyperbola having eccentricity as ‘e’ and C be the smallest circle touching H and its pair of asymptotes. Ratio of radius of C with the length of semi-conjugate axis of H equals

(A) e
(B) $e+\sqrt{e^2-1}$
(C) $e-\sqrt{e^2-1}$
(D) $\frac{1}{e}$

39. Let the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ and the hyperbola $\frac{x^2}{A^2}-\frac{y^2}{B^2}=1$ be such that their respective end points of latus rectum coincide, then sum of squares of their eccentricities equal

(A) $\frac{7}{4}$
(B) 2
(C) $\frac{11}{4}$
(D) None of these

40. If the lines 2x + y + c = 0 and x + 2y + c = 0 ; c = 1, 2, 3 intersect the hyperbola H at exactly one point each, then the eccentricity of H can be equal to

(A) $\frac{5}{4}$ (B) $\sqrt{2}$

(C) $\frac{\sqrt{10}}{3}$ (D) none of these

41. Number of solutions of the equation $f(x) + x - \pi = 0$ where $f(x) = \max\{\sin x, \cos x\}$ is

(A) 0 (B) 1

(C) 3 (D) 5

SECTION – A

(Matching List Type)

*This section contains **FOUR (04)** Matching List Type Questions. Each question has FOUR statements in **List-I** entries (P), (Q), (R) and (S) and FIVE statements in **List-II** entries (1), (2), (3), (4) and (5). The codes for lists have choices (A), (B), (C), (D) out of which **ONLY ONE** of these four options is correct answer.*

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42. Let $\sin^{-1}x : [-1, 1] \to \left[\frac{5\pi}{2}, \frac{7\pi}{2}\right]$ and $\cos^{-1}x : [-1, 1] \to [2\pi, 3\pi]$, then match number of integers in range of functions in List-I with List-II

LIST – I		LIST - II	
(P)	$\frac{1}{\pi}\left(\sin^{-1}x + \cos^{-1}x\right)$	(1)	0
(Q)	$\frac{1}{\pi}\left(\sin^{-1}x - \cos^{-1}x\right)$	(2)	1
(R)	$\frac{1}{\pi^2}\left(\sin^{-1}x \cdot \cos^{-1}x\right)$	(3)	2
(S)	$\cos^{-1}\left(\frac{\sin^{-1}x}{5\pi}\right)$	(4)	4
		(5)	6

The correct option is:

(A) (P) $\to$ (1); (Q) $\to$ (3); (R) $\to$ (1); (S) $\to$ (2)
(B) (P) $\to$ (1); (Q) $\to$ (3); (R) $\to$ (1); (S) $\to$ (1)
(C) (P) $\to$ (3); (Q) $\to$ (1); (R) $\to$ (5); (S) $\to$ (1)
(D) (P) $\to$ (2); (Q) $\to$ (1); (R) $\to$ (5); (S) $\to$ (1)

43. Let A + B + C = π and $(\sin^2 A + \sin^2 B - \sin^2 C)^2 = p$, where range of p is [q, r]. Let DEF be a triangle and $\tan^2 F (\sin^2 D + \sin^2 E - \sin^2 F)^2 = p'$ where range of p' is $(q', r']$, then match List-I with List-II and select the correct answer using the codes given below the list:

LIST – I		LIST - II	
(P)	rr'	(1)	0
(Q)	$rq - r'q'$	(2)	$\frac{27}{16}$
(R)	$r + q + 4r' + q'$	(3)	$\frac{27}{4}$

(S)	$\frac{16r'}{r}$	(4)	27
		(5)	$\frac{43}{4}$

The correct option is:
(A) (P) → (2); (Q) → (1); (R) → (3); (S) → (3)
(B) (P) → (3); (Q) → (1); (R) → (5); (S) → (3)
(C) (P) → (4); (Q) → (1); (P) → (2); (S) → (4)
(D) (P) → (3); (Q) → (1); (R) → (4); (S) → (4)

44. Let L : (l + pm)x + (m + pl)y + n + np = 0 be a variable line where l, m, n are positive constants and p be a parameter. Then match statements/expression in List-I with corresponding values in List-II

LIST – I		LIST - II	
(P)	If maximum possible distance of L from (2, 2) is less than $3\sqrt{2}$ units, then l + m – n can be equal to	(1)	0
(Q)	Number of possible triangles that L can form ($\forall$ p $\in$ R) with coordinate axes having area equal to lmn sq. units if L passes through (2, 2) is	(2)	1
(R)	Number of values of p for which L is a tangent to $(l + m)^2(x^2 + y^2) - 3n^2 = 0$ is	(3)	2
(S)	Number of values of p for which L can be an angle bisector of coordinate axes is	(4)	3
		(5)	∞

The correct option is:
(A) (P) → (1); (Q) → (5); (R) → (3); (S) → (2)
(B) (P) → (4); (Q) → (1); (R) → (1); (S) → (2)
(C) (P) → (2); (Q) → (4); (R) → (3); (S) → (5)
(D) (P) → (3); (Q) → (5); (R) → (3); (S) → (1)

45. Let $S_1 : x^2 + y^2 - 2x - 2y + 1 = 0$;
$S_2 : x^2 + y^2 = 9$;
$S_3 : x^2 + y^2 - 6\sqrt{2}x - 6\sqrt{2}y + 27 = 0$ and
L be common tangent to S_2 and S_3, then match locus of point in List-II with List-I

LIST – I		LIST - II	
(P)	Locus of centre of circle touching both S_1 and S_3 externally is a	(1)	Line
(Q)	Locus of centre of circle touching both S_2 and S_3 externally is a	(2)	Parabola
(R)	Locus of centre of circle touching S_1 externally and S_2 internally is a	(3)	Ellipse
(S)	Locus of centre of circle touching S_1 and L is a	(4)	Hyperbola
		(5)	None

The correct option is:
(A) (P) → (5); (Q) → (1); (R) → (3); (S) → (2)
(B) (P) → (4); (Q) → (1); (R) → (5); (S) → (2)
(C) (P) → (4); (Q) → (1); (R) → (3); (S) → (5)
(D) (P) → (4); (Q) → (5); (R) → (3); (S) → (2)

SECTION – B

(Numerical Answer Type)

*This section contains **SIX (06)** Numerical based questions. The answer to each question is a **NON-NEGATIVE INTEGER VALUE**.*

46. Number of solutions of the equation $\sin^{-1}(x-1)^2 + \cos^{-1}\left(\frac{1}{x^2-1}\right) = \tan^{-1}x$ is (considering only the principal values of inverse trigonometric function)

47. Number of solutions of the set of equations $\sin\theta = \cos 5\theta$ and $\tan 2\theta = \cot 4\theta$ $\forall\ \theta \in [0, 2023\pi] - \left\{\frac{(2n+1)\pi}{8}\right\} - \left\{\frac{(2p+1)\pi}{4}\right\}$; n, p $\in$ I is

48. Let $\tan\alpha$, $\tan\beta$, $\tan\gamma$ be solutions of the equation $\sqrt{3}x^3 - 3x^2 - 3\sqrt{3}x + 1 = 0$ where $\alpha, \beta, \gamma \in [0, \pi]$ $\alpha < \beta < \gamma$, then $\frac{24}{\pi}$ $(\alpha + \beta + \gamma)$ equals

49. If the angles of a triangle are consecutive terms of an Arithmetic Progression as well as a Geometric Progression, then sum of all possible value(s) of the square of ratio of semi-perimeter of the triangle with its in radius equals

50. Minimum value of the expression $(x_1 - x_2)^2 + \left(3 + \sqrt{4x_1} - \sqrt{1-x_2^2}\right)^2$ for all permissible value(s) of x_1, x_2 is p, then [p] equals (where [.] represents greatest integer function)

51. Let H : $x^2 - y^2 = 1$ and A be its pair of asymptotes. Let P be a parabola having the same vertex as H and having A as tangents. Area bounded by latus rectum of P with A equals (in sq. units)